Retail banking warehouse prototype

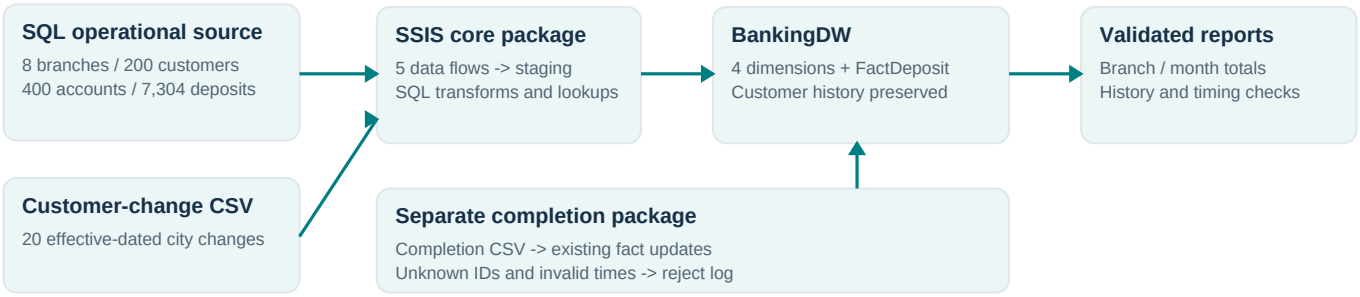
Working SQL Server and SSIS delivery | Technical handover

7,304	365	4	100%
deposit facts	calendar days	dimensions	facts completed

182,891,102.97 synthetic currency units across 8 branches, 200 customers and 400 accounts. The original four sample deposits are preserved inside the generated year.

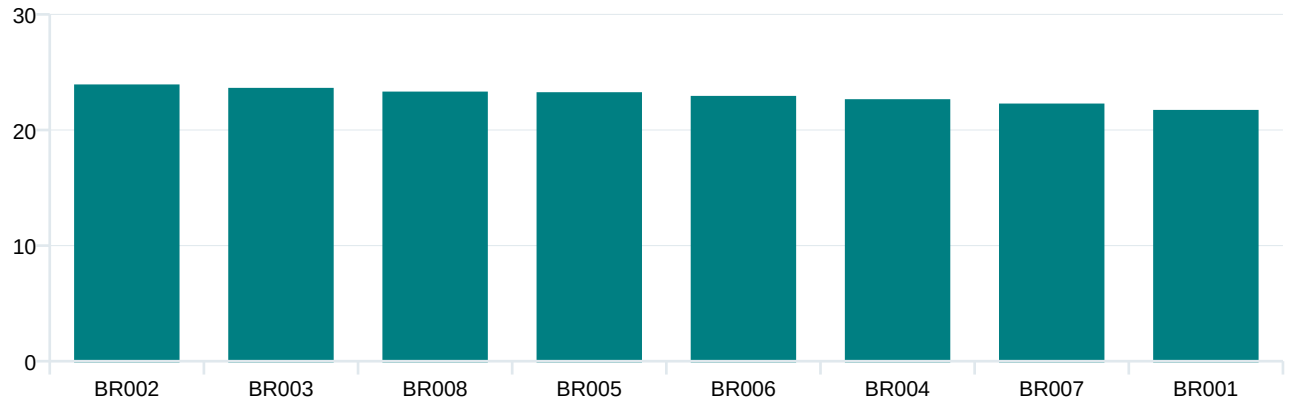
This is a functioning local prototype using fictional records. The figures demonstrate analytical capability; they are not observations about a real bank.

Implemented flow



A deposit fact is one transaction. The handling branch is kept separate from the account home branch. Customer versions are selected using the deposit date.

Example analytical output



Handling-branch deposit totals, millions of synthetic currency units.

Demo Maharagama (BR002) has the highest generated total: 23,825,484.11, or 13.03% of the scenario total.

1,819 deposits (24.9%) were handled away from the account home branch. This demonstrates why those two branch roles must remain distinct.

Daily volume is generated at 20 deposits per day, plus the original fixture. Monthly volume differences therefore partly reflect calendar length, not observed business growth.

Delivery evidence and handover

The following checks were run against the actual SQL Server warehouse and native SSIS packages.

Validation	Result
Seven automated source tests: keys, money, dates, repeatability and history inputs	PASS
Every operational source field reconciled with generated CSV records	PASS
Source and fact IDs, counts and amount totals reconcile	PASS
Customer intervals do not overlap; facts use the correct dated version	PASS
Both SSIS reruns preserve fact keys, values and timestamps (SHA-256 check)	PASS
Unknown IDs and completion-before-creation inputs are rejected without changing valid facts	PASS

Historical city example: customer C001

Deposit period	Customer city	Deposit facts
1 Jan - 30 Jun 2026	Piliyandala	32
From 1 Jul 2026	Maharagama	41

Twenty customer city changes produce 220 dimension versions for 200 customers. Earlier facts retain their original city context.

Concrete engineering work

Source engineering: fixed-seed Python generator, exact decimal validation, relational checks and safe repeatable loading.

Warehouse design: a star schema with surrogate keys, a Date dimension and Type 2 Customer history.

Integration: two real SSIS packages, SQL and Unicode CSV data flows, dimension lookups, fact upserts and a separate completion update.

Delivery: automated tests, SQL assertions, audit/reject tables, reusable report queries and a single rerun command.

Run the complete demonstration

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File .\ssis\run_demo.ps1
```

The workflow regenerates sources, runs seven tests, loads both SSIS packages, validates the warehouse and exports branch/month report data. Detailed reproduction and the schema are in WAREHOUSE_HANDOVER.md.

Scope and next work

The warehouse core is implemented and validated locally. Completion hours (0.5 to 3.25) are deliberately simulated from warehouse creation time; they are not measured customer-service SLAs.

SSAS/Excel/Power BI presentation work, production deployment and a future lakehouse adaptation remain separate next steps. No real customer records or confidential workplace data are used.